

INTRODUCTION

Rice (*Oryza sativa*) is a vital staple crop in the Philippines, with the country being one of the top rice producers in the world. Rice cultivation not only sustains national food security but also supports the livelihoods of millions of Filipino farmers, as approximately 80 percent of the population relies on rice as a primary dietary source (Vanzi, 2023). While rice in the Philippines is grown throughout the year, the largest production share is grown during the wet season. Due to this diversity of planting and harvesting, the government of the Philippines takes annual, semester, and quarterly statistics on rice production and harvested areas. Farms in the Philippines are generally small (less than two hectares on average, which may limit the implementation of advanced farming technologies). Currently, irrigated paddy rice accounts for 60% of total production, with the remainder grown as upland directly seeded rice (Stuecker, 2018).

Despite its importance, rice production faces ongoing challenges from pests, particularly birds that feed on developing and mature grains. They also face environmental field problems disrupting their growth rate and quality. Species such as the Maya (*Passer montanus*) and Mayang pula (*Lonchuramalacca*) cause significant crop losses by chewing grains during the milky phase or consuming fully matured rice, resulting in unfilled panicles or whiteheads (Tababa, 2023; IRRI Knowledge Bank, n.d.). Rice cultivation requires large amounts of water, and farmers in drought-prone areas can struggle to secure enough water for their crops. Crop yields are strongly affected by weather and water availability. Greater irrigation is needed during sunny and dry seasons, while cold and wet seasons can lead to reduced photosynthesis, disease, and grain sterility (Rajan, 2025).

Traditional methods for bird deterrence such as scarecrows, flags, kites, and noisemakers offer only temporary relief, as birds often adapt to these stimuli over time (IRRI Rice Knowledge Bank, n.d.). The array of visual bird deterrents is expansive, and includes lights that are flashing or rotating, searchlights, mirrors and reflectors, reflective tape, flags, rags, streamers, lasers, dogs, humans, scarecrows, raptor models, corpses, balloons with eyespots, kites, kite hawks, mobile predator models, and water dyes or colorants (Bishop et al. 2003). All these methods work to some degree for a short period of time until habituation. Lasers that were used to disperse crows, for example, resulted in an initial dispersion, but crows reoccupied their roosts the same night that the lasers were used, and none of the roosts were abandoned (Gorenzel et al. 2002). During drought events, vegetation canopies can be affected by water stress. This can have major impact on the plant development in general and can cause crop failure or lower crop production in agricultural areas. Early recognition of plant water stress can be critical to prevent such consequences (de Lima, 2021).

To address these limitations, this study proposes the development of the device is designed to integrate automated bird detection with real-time monitoring of water levels and soil moisture, providing timely notifications to farmers via SMS and a web-based dashboard. By combining pest management with environmental monitoring in a single platform, the device aims to enhance crop protection, optimize irrigation, and reduce labor-intensive manual supervision, thereby improving efficiency and productivity for rice farmers.

Despite prior advancements in automated bird deterrence, few existing systems integrate environmental monitoring with pest management in an affordable, farmer-accessible solution. Many devices remain expensive, technologically complex, or limited in usability, leaving small-scale farmers with suboptimal tools for crop protection. This device seeks to bridge this gap by

offering a cost-effective, user-friendly, and multi-functional device that addresses both pest control and irrigation management while delivering timely notifications for proactive decision-making.

This study focuses on the design, implementation, and evaluation of the device under field conditions in rice paddies. It aims to assess the device's effectiveness in repelling birds, the accuracy and reliability of water level and soil moisture measurements, the timeliness of notifications to farmers, and the overall usability and acceptability of the system. By integrating automated monitoring, real-time alerts, and energy-efficient operation, this provides a practical, sustainable, and technologically advanced solution to longstanding challenges in Philippine rice farming, ultimately contributing to improved productivity, reduced crop losses, and strengthened agricultural resilience.

STATEMENT OF THE PROBLEM

This study aims to develop a smart agricultural monitoring and bird deterrence system that can detect bird activity in real time, monitor soil moisture and water level conditions using distributed sensors, and send alerts and data through SMS and a mobile application interface to improve crop protection, field monitoring efficiency, and farming management.

Specifically, this study seeks to answer the following questions:

- 1) What is the accuracy of bird detection using the Raspberry Pi camera with YOLO algorithm?
- 2) What is the response time of the system in detecting birds in real time?
- 3) How effective is the automated auditory mechanism in deterring bird activity?
- 4) How accurately does the system monitor field conditions, specifically:
 - a) Water level
 - b) Soil moisture
- 5) How efficient is the system in processing and transmitting sensor data?
- 6) How reliable and timely is the system in sending SMS alert notifications for critical condition to users?
- 7) How effective is the mobile application in displaying real-time data and camera feed?
- 8) What are the users' insights on the AGRI-BIRD device in terms of:
 - a) Functionality
 - b) Reliability
 - c) Usability
 - d) Portability
 - e) Maintainability
 - f) Accuracy

HYPOTHESIS

Null Hypothesis:

The use of the RICE-GUARD system has no significant effect on reducing bird activity in rice fields and on improving the accuracy and efficiency of irrigation monitoring compared to traditional methods.

Alternative Hypothesis:

The use of the RICE-GUARD system significantly reduces bird activity in rice fields and improves the accuracy and efficiency of irrigation monitoring compared to traditional methods.

ENGINEERING GOALS

The primary goal of this study is to design and develop an automated, solar-powered agricultural monitoring and bird deterrent system. This study aims to Develop a vision-based detection system using Raspberry Pi and YOLO for accurate bird identification. Design an automated auditory deterrent mechanism that reduces bird presence using randomized predator sounds. Integrate soil moisture and water level sensors for real-time environmental monitoring. Implement a GSM-based SMS notification system for timely alerts to farmers.

EXPECTED OUTCOME

This study aims to provide a solution in deterring bird activity to protect rice fields, and monitors soil moisture and irrigation. The device is expected to achieve high accuracy in detecting birds using the YOLO algorithm, the response time between detection and deterrent activation will be fast and consistent, the auditory deterrent mechanism will significantly reduce bird presence in the field, soil moisture and water level sensors will provide accurate and reliable environmental data, SMS notifications will be delivered in real time with minimal delay, and the

METHODOLOGY

This study will use an experimental method and design in determining the effectiveness, acceptability, efficiency, and functionality of the device.

Collection and Preparation of Materials

The researcher will purchase the materials necessary for the construction of the system through both online purchasing platforms and local hardware stores. The materials including the Raspberry Pi 4, camera module, GSM module, relay module, microcontrollers, capacitive soil moisture sensors, water level sensors, speaker units, voltage regulators, jumper wires, breadboards, and communication modules will be purchased online to ensure availability and compatibility of electronic components.

Other materials such as electrical wires, connectors, mounting poles, protective casings, fasteners, and structural components for the scarecrow framework will be obtained from local hardware and electronics stores around Polangui, Albay. Additional tools including a soldering iron, solder wire, and electrical tape will also be acquired locally for assembly and installation purposes.

Central and Structural Integration Unit

The main device will be a multi-functional structural unit designed to resemble a humanoid scarecrow, serving both as a computational hub and a physical bird deterrent. Rather than functioning as a purely symbolic structure, this unit will be engineered to integrate sensing, processing, and actuation components within a single elevated platform.

The unit will be strategically installed at a corner of the agricultural field to achieve maximum observational coverage and efficient signal distribution, while minimizing interference

with farming operations. Its elevation, mounted by a reinforced vertical support pole, will improve line-of-sight for visual detection and will enhance the propagation range of deterrent signals. The structure will be divided into three integrated functional compartments:

1) Head Section (Visual Acquisition Unit):

This section will contain the camera module mounted at an optimal height and angle to capture a wide field of view. It will continuously stream real-time video to the central processor, enabling bird detection through image analysis techniques such as motion tracking and object recognition.

2) Torso Section (Processing and Control Unit):

Located within the main body, this compartment will house the Raspberry Pi 4, which will serve as the primary controller of the system. Supporting components will include the relay module, voltage regulators, and thermal management elements such as heat sinks or cooling fans. All electronics will be enclosed in a weather-resistant casing to ensure protection against environmental factors such as rain, dust, and heat, while maintaining proper ventilation.

3) Arm Section (Actuation and Output Unit):

Extending from the main structure, this section will support the dual speaker systems positioned on opposite ends. These speakers are responsible for emitting deterrent sounds upon detection of bird activity. Their horizontal placement ensures wider acoustic coverage that will sound waves to propagate evenly across the monitored area.

The design also will reinforce the deterrent effect by combining visual presence in a scarecrow form with automated auditory response, creating a multi-sensory approach to bird control.

Plate 1: Shows the Labeled Structural Design of the Central Unit

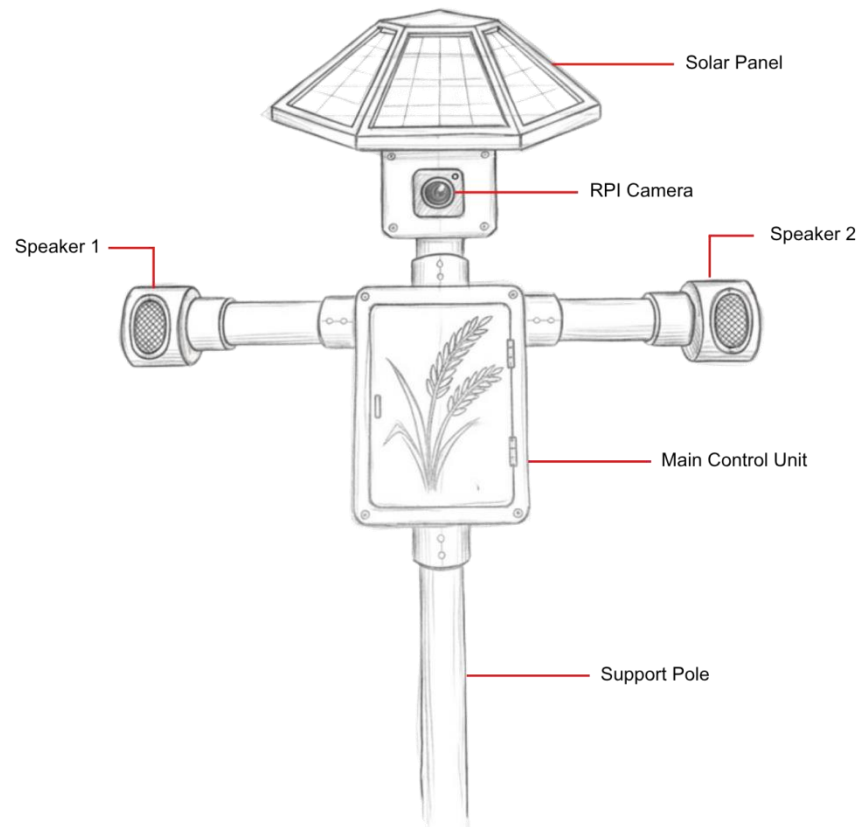
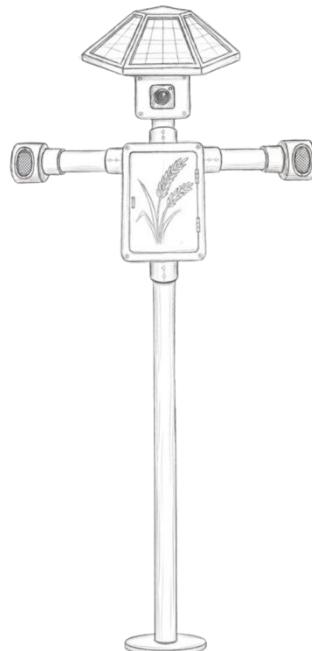


Plate 2: Shows the Full Front View of the Central Unit



Bird Detection and Deterrence Mechanism

The system will incorporate a real-time computer vision-based detection and automated deterrence mechanism designed to identify and respond to bird activity within the agricultural field. The detection process will be implemented using the You Only Look Once (YOLO) object detection algorithm, deployed through the Python programming environment and integrated with the central processing unit.

The YOLO model will be trained using a dataset composed of bird images captured under varying environmental conditions, including differences in lighting, orientation, background complexity, and distance. The training process will focus on enabling the model to accurately recognize bird presence across multiple spatial ranges within the field of view.

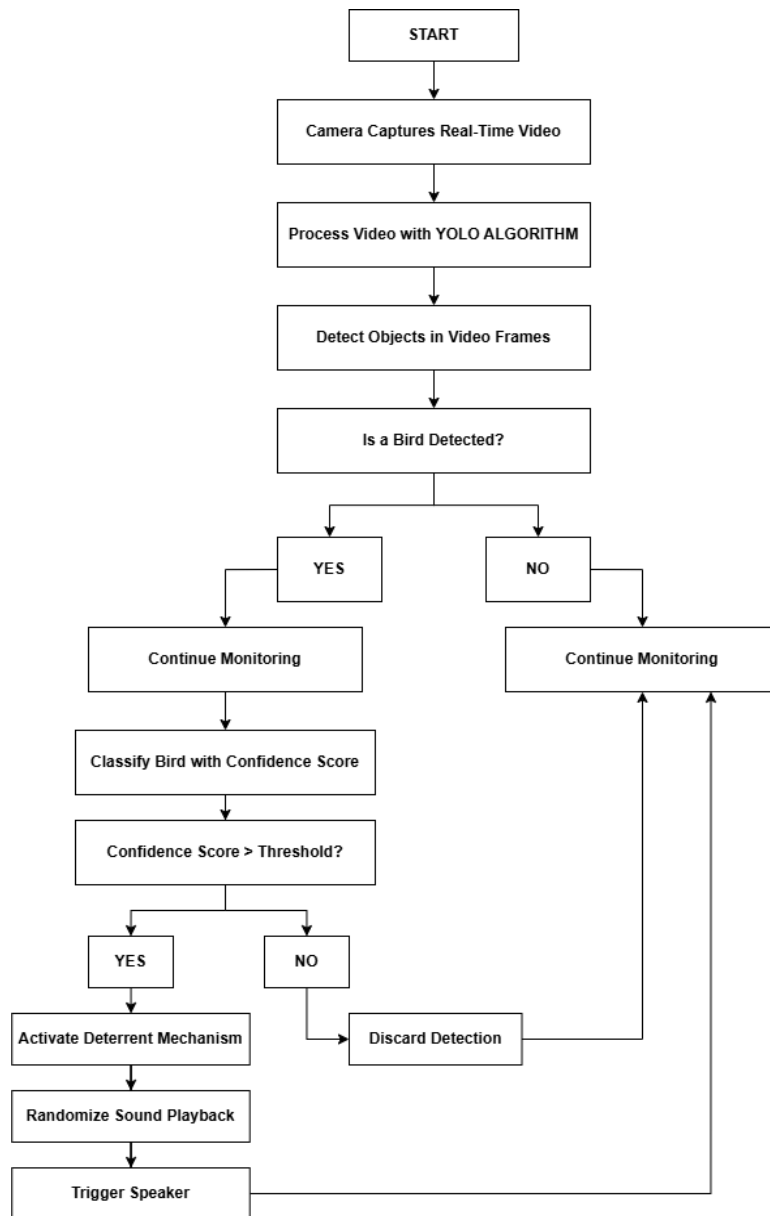
During operation, the camera will continuously capture image frames which will be processed in real time by the trained YOLO model. Detected objects will be classified and assigned confidence scores, and only those exceeding a predefined threshold will be considered valid detections. This filtering mechanism will reduce false positives caused by non-target movements such as vegetation motion or lighting changes.

Once bird presence is confirmed, the system will activate the deterrence mechanism through a relay-controlled audio output system. Two speakers mounted at the horizontally extended arms of the structure will be triggered simultaneously to produce directional sound propagation across the field.

The audio output will consist of preloaded predator calls and bird distress signals stored within the system. To prevent habituation, a randomization strategy will be implemented in the selection and timing of sound playback. Variations in sound type and activation intervals will be applied to maintain unpredictability and sustain the effectiveness of the deterrence response.

The detection and deterrence processes will operate in a continuous loop, ensuring that bird activity will be identified and addressed in real time.

Figure 1. *Bird Detection Process Flowchart*



Distributed Sensor Network Design

Environmental monitoring will be implemented through a distributed sensor network arranged in a grid-based layout across the agricultural field. This layout will be designed to

capture spatial variations in soil and water conditions, especially in areas with uneven terrain or differing moisture retention levels.

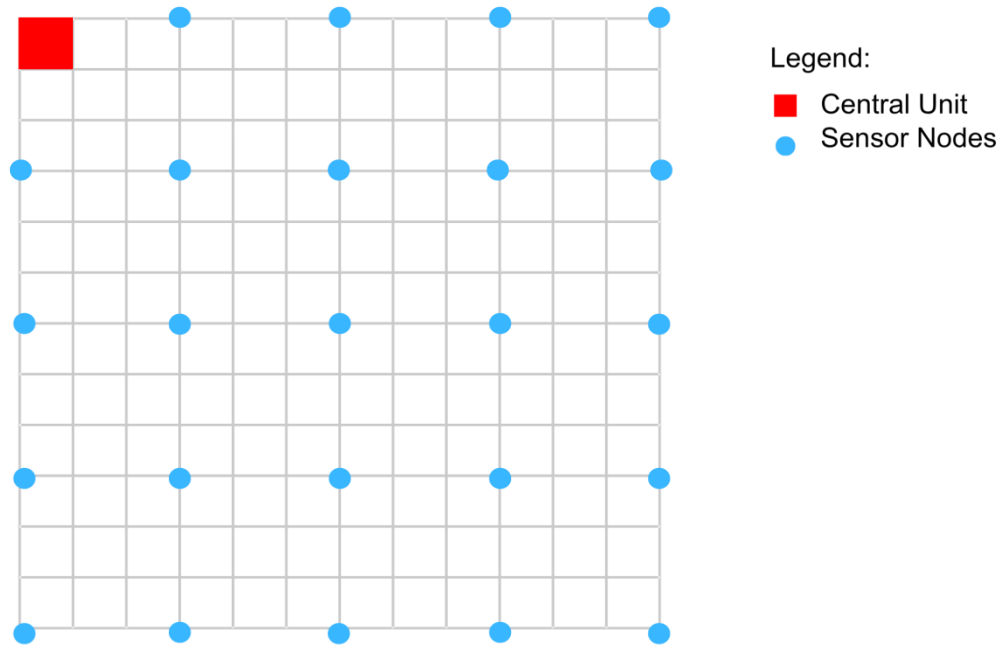
Each sensor node will be installed on a fixed vertical mounting pole anchored into the ground, ensuring stability and protection from field activities such as plowing, irrigation, and foot traffic. The elevated positioning will also help reduce direct exposure to surface water, mud accumulation, and accidental impact from farming tools. Each node will be composed of the following components:

- A **capacitive soil moisture sensor** for measuring volumetric water content in the soil
- A **water level sensor** for detecting surface or accumulated water conditions
- A **low-power microcontroller unit** for local data handling
- A protective casing to shield components from rain, dust, and heat exposure

Unlike traditional systems that continuously transmit raw sensor data, each node will function as a semi-autonomous sensing unit capable of performing initial data processing before transmission.

This distributed arrangement will allow the system to capture localized environmental differences, which will be especially important in agricultural fields where soil conditions can vary significantly even within short distances. The grid structure will also ensure that no area will be left unmonitored. Nodes closer to drainage zones or low-lying areas may provide more critical readings for flood detection, while elevated zones will focus more on soil moisture variability.

Figure 2. *Top-View Grid-Based Sensor Network Layout*



Node-Level Data Processing and Aggregation

At the sensor node level, data handling will be performed locally before any transmission to the central system. This approach will reduce unnecessary communication load and will ensure that only meaningful environmental information will be forwarded for higher-level analysis.

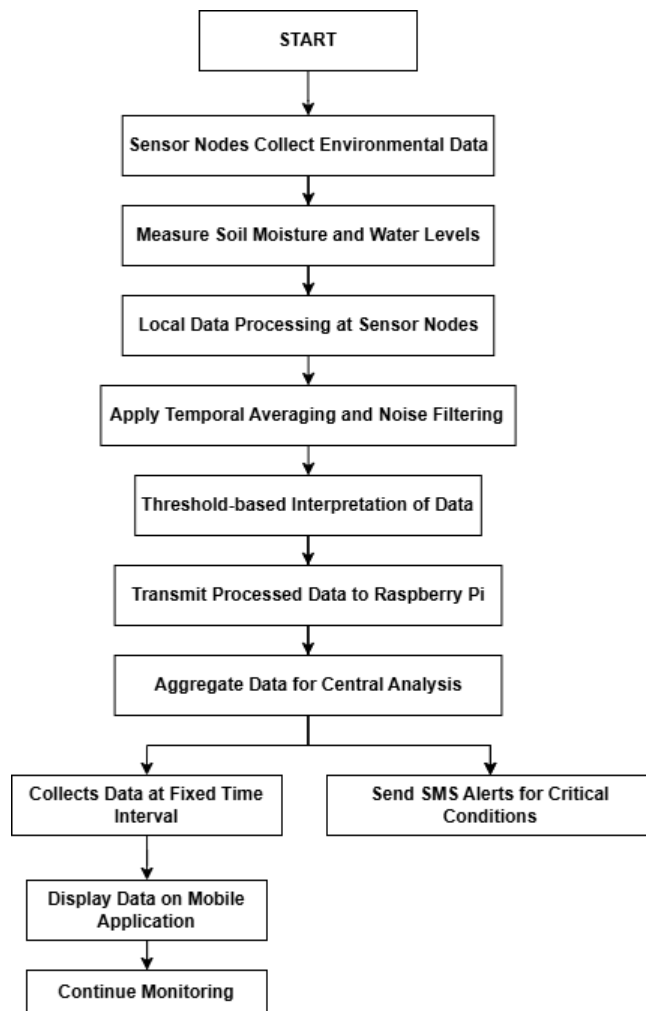
Each sensor node will continuously collect readings from the soil moisture and water level sensors at fixed time intervals. Instead of sending every individual reading, the node will store a short sequence of measurements and will apply basic pre-processing techniques to improve reliability and reduce noise. These techniques will include;

- **Temporal averaging:** Multiple readings collected within a set time window will be averaged to smooth out sudden fluctuations caused by sensor instability or transient environmental disturbances.
- **Noise filtering:** Abnormal spikes or inconsistent values will be reduced or discarded using simple rule-based filtering, ensuring more stable output values.
- **Threshold-based interpretation:** Raw sensor values will be converted into categorized conditions to simplify transmission and analysis.

For soil moisture, the processed output will be represented as a representative moisture level for a given interval. For water level sensing, values will be interpreted into predefined states such as low, moderate, or critical.

After processing, each node will transmit only the summarized and classified data to the Raspberry Pi. This will significantly reduce the volume of transmitted information compared to raw-data streaming systems.

Figure 3. *Environmental Monitoring Process Flowchart*



Communication and Wiring System

The sensor nodes will be interconnected with the central unit through an elevated wired communication network installed above crop level. The wiring system will be arranged along supporting poles and will be routed in an organized manner to maintain structured connectivity across the agricultural field.

The elevated configuration will minimize exposure of cables to soil contact, water accumulation, and mechanical disturbance caused by routine farming activities. The communication design will ensure stable data transmission from distributed nodes to the central

unit while maintaining physical separation from field operations and reducing the risk of hardware damage.

Mobile Application Interface

The mobile application will be developed as the primary user interface of the system, enabling remote monitoring and basic control of the entire agricultural setup. It will communicate wirelessly with the Raspberry Pi using a network connection.

This application will receive processed information from the central unit rather than raw sensor data, ensuring that displayed values are already filtered, summarized, and system-ready. It will act strictly as a monitoring and command interface, while all analysis and decision-making processes will remain centralized within the Raspberry Pi system.

The main features of the mobile application will include:

- **Real-time environmental monitoring:** It will display aggregated soil moisture levels and water condition classifications from all sensor nodes in the field. Data will typically be presented in a simplified format, allowing users to quickly identify dry, optimal, or waterlogged areas.
- **Live video streaming:** It will provide direct access to the camera feed from the central scarecrow unit, functioning similarly to a CCTV system for continuous field observation and bird activity monitoring.
- **Basic system control functions:** It will allow users to enable or disable system operations and manage key functions such as activating monitoring mode or resetting system states.

GSM-Based Alerts Notification System

The system will incorporate a GSM-based communication module integrated with the Raspberry Pi 4 only for SMS-based alert transmission compared to the mobile interface that provides data in a time interval. This module will operate as a standalone notification channel and does not require internet connectivity, enabling functionality in offline or low-network-coverage environments.

The GSM module will only be triggered by predefined conditions processed at the central unit. Alert conditions will include:

- **Bird detection event:** SMS will be generated when the vision system detects bird presence above a defined threshold.
- **Soil moisture threshold condition:** SMS will be sent when soil moisture readings fall below or exceed configured limits indicating dry or saturated soil conditions.
- **Water level threshold condition:** SMS will be triggered when water level readings indicate abnormal states such as flooding or insufficient water availability.

The module will enable direct transmission of SMS notifications to registered recipients, allowing remote users to receive real-time system status updates without relying on internet-based communication platforms.

The system will also allow registered users to send a predefined command (e.g., “STATUS”) to the GSM module, prompting the central processing unit to return an SMS containing the current processed and summarized data, including bird detection activity, average soil moisture level, and water level status.

Connections and Integration

The Raspberry Pi 4 will act as the main processing unit of the system. The connections of this device to the system components are shown in Table 1.

Table 1: Connections of the Raspberry Pi 4 to System Components

Component	Connection
Camera Module	CSI Port
Sensor Node Network	GPIO / UART
Relay Module	GPIO Output Pin
Mobile Communication Module	Wi-Fi / UART
Power Supply	5V and GND

The Raspberry Pi 4 will receive aggregated data from the sensor nodes and process real-time video from the camera module. It will also control the relay module to activate the speaker system and manage communication with the mobile application for monitoring and system control.

The Camera Module will capture real-time video of the agricultural field for bird detection and monitoring. The connections of this module to the Raspberry Pi 4 are shown in Table 2.

Table 2: Connections of the Camera Module to the Raspberry Pi 4

Camera Module	Connection
VCC	CSI Power Line
GND	CSI Ground
Data	CSI Ribbon Interface

The camera module will continuously stream visual data to the Raspberry Pi 4. This will allow the system to detect bird presence through motion analysis and object recognition.

The Soil Moisture Sensor will measure the water content of the soil across different field locations. The connections of this sensor to the node microcontroller are shown in Table 3.

Table 3: Connections of the Soil Moisture Sensor to Node Microcontroller

Soil Moisture Sensor	Connection
VCC	5V
GND	GND
Analog Output	ADC Input Pin

The sensor will collect soil moisture readings which will be processed locally at the node level. The processed values will then be transmitted to the central system.

The Water Level Sensor will detect surface water conditions in the field. The connections of this sensor to the node microcontroller are shown in Table 4.

Table 4: Connections of the Water Level Sensor to Node Microcontroller

Water Level Sensor	Connection
VCC	5V
GND	GND
Signal Output	Digital/Analog Input Pin

The sensor will classify water conditions such as normal, saturated, or flooded. The processed data will be sent to the Raspberry Pi 4 for analysis.

The Sensor Nodes will transmit processed environmental data to the Raspberry Pi 4 through wired communication. The connections of this network are shown in Table 5.

Table 5: Connections of Sensor Nodes to Raspberry Pi 4

Sensor Node	Raspberry Pi 4
TX	RX (UART/GPIO)
RX	TX (UART/GPIO)
GND	GND

The sensor nodes will send only summarized data such as average soil moisture and water level status to reduce data redundancy and improve efficiency.

The Relay Module will control the activation of the speaker system used for bird deterrence. The connections of this module to the Raspberry Pi 4 are shown in Table 6.

Table 6: Connections of the Relay Module to the Raspberry Pi 4

Relay Module	Connection
VCC	5V
GND	GND
IN	GPIO Output Pin

The relay module will act as a switching interface that allows the Raspberry Pi 4 to safely control the speaker system.

The Speaker System will produce sound-based deterrent signals when activated. The connections of this system to the relay module are shown in Table 7.

Table: Connections of the Speaker System to the Relay Module

Speaker System	Connection
Positive (+)	COM Output
Negative (-)	External Ground

The speaker system will emit deterrent sounds when bird activity is detected, helping to protect crops from damage.

The Power Supply System will provide energy to all components of the system. The connections of this system are shown in Table 8.

Table 8: Connections of the Power Supply System

Power Component	Connection
Solar Panel	Charge Controller
Charge Controller	Battery

Battery	Voltage Regulator
Voltage Regulator	All System Components

The power system will ensure stable and continuous operation of the entire system using regulated energy distribution.

The Mobile Application and SMS Notification System will provide remote monitoring, control, and alert delivery for the system. The connections of this module to the Raspberry Pi 4 are shown in Table 9.

Table 9: Connections of the Interfaces to the Raspberry Pi 4

Interface	Communication Method	Connection
Mobile Application	Wi-Fi / Hotspot	Wireless Network
SMS Notification System	GSM Module	UART Pins

The mobile application will be used for real-time monitoring and control of the system. It will display live sensor readings, system status, and camera feed, allowing users to remotely observe field conditions.

The SMS notification system will deliver critical alerts such as bird detection events and abnormal sensor. These alerts will be transmitted to designated users even without internet access, ensuring continuous communication in remote agricultural areas

DATA GATHERING AND ANALYSIS

Assessment of Bird Detection (YOLO Algorithm)

The accuracy of bird detection will be evaluated through the deployment of the device in a controlled agricultural environment where known bird species are present. The device will continuously capture video data over a specified period, allowing for the collection of real-time detection instances. Concurrently, trained observers will conduct manual observations to establish a baseline for actual bird occurrences, which will serve as the ground truth against which the device's detections will be compared.

The footage will be analyzed using the Raspberry Pi Camera trained with YOLO (You Only Look Once) algorithm, a sophisticated object detection system recognized for its speed and accuracy. The algorithm will process video frames in real time to detect birds, logging instances along with timestamps. Detected instances will be categorized as true positives, false positives, or false negatives based on the ground truth data.

To analyze the results, a confusion matrix will be constructed to summarize the performance of the bird detection algorithm. This matrix will provide a clear visual representation of the detection outcomes, detailing counts of true positives, false positives, true negatives, and false negatives. The Key Evaluation Metrics to be used are:

- 1) **Accuracy:** The proportion of correctly predicted instances (true positives and true negatives) relative to the total instances.

$$\text{Accuracy} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives} + \text{False Negatives}}$$

- 2) **Precision:** The proportion of true positive detections to the total positive detections.

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives}$$

- 3) **Recall (Sensitivity):** The proportion of true positive detections to actual positives.

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives}$$

Where:

- *True Positives* (TP) is the number of correctly detected instances of a class
 - *False Positives* (FP) is the number of instances incorrectly identified as belonging to a class.
 - *False Negatives* (TN) is the number of instances that belong to a class but were not detected.
- 4) **F1-Score:** The harmonic mean of precision and recall, providing a balanced measure of model performance.

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

- 5) **Mean Average Precision (mAP):** Evaluates precision and recall across multiple classes.

$$mAP = \frac{1}{C} \sum_{i=1}^C AP_i$$

Where:

- C is the total number of classes
- AP_i is the Average Precision for class i
- P is Precision

6) Receiver Operating Characteristic (ROC) curve:

A graphical plot that illustrates the trade-off between sensitivity (true positive rate) and specificity (true negative rate) across different threshold settings. The area under the ROC curve (AUC) will serve as a single measure of overall accuracy, facilitating comparisons across different models or configurations

Table 1: *Bird Detection Results*

	Trial										Total
	1	2	3	4	5	6	7	8	9	10	
True Positives (TP)											
False Positives (FP)											
False Negatives (FN)											
Accuracy (%)											

System Response Time in Real-Time Detection

The response time of the system will be evaluated by measuring the interval between the moment a bird is detected and the subsequent activation of the audio deterrent. This measurement is critical, as a rapid response time may enhance the device's effectiveness in deterring birds and thereby reducing potential crop damage.

An automated logging system will be implemented to accurately record timestamps of detection events and audio activation. This system will ensure high precision in timing,

minimizing human error. A series of detection events will be simulated, targeting a minimum of 50 detection instances to ensure reliability in the findings.

Each event will involve a bird being detected by the camera, followed by the system activating the audio deterrent. The time intervals between detection and activation will be recorded, providing a comprehensive dataset for analysis. Descriptive statistical methods will be utilized to summarize the system's response times, including measures of central tendency and variability.

Statistical measures will be computed to provide a comprehensive overview of the data, specifically:

- Mean: $\bar{X}$
- Median: $\tilde{X}$
- Standard Deviation: S

Subsequently, a 95% confidence interval will be calculated to assess the reliability and precision of the estimated mean response time, expressed as:

$$CI = \bar{x} \pm z \frac{s}{\sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
- $\tilde{x}$ = median
- S = standard deviation
- n = sample size
- Z = z-score corresponding to 95% confidence level

Table 2: Response Time Data

	Trial										Average
	1	2	3	4	5	6	7	8	9	10	
Detection Time (hh:mm:ss)											
Audio Activation Time (hh:mm:ss)											
Response Time (seconds)											

Effectiveness of the Automated Auditory Mechanism in Deterring Bird Activity

To assess the effectiveness of the automated auditory mechanism in deterring bird activity, a field study will be conducted over a predetermined period of two weeks. This study will involve monitoring bird activity in the presence and absence of the device, utilizing video surveillance to track bird presence and behavior before and after the activation of the audio deterrent.

The number of birds per hour will be quantified and recorded during both phases of the study. Data will be collected in both scenarios (with and without the device) to facilitate a comparative analysis of the device's impact on bird activity levels.

Statistical analyses will be conducted to compare the system's performance under different conditions. A t-test will be utilized to determine whether there is a significant difference between the mean response times before and after device activation. If comparisons involve more than two groups, such as varying times of day or different environmental conditions,

analysis of variance (ANOVA) will be employed to assess overall differences among group means. The t-test statistic is expressed as:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{2}{n}}}$$

Where:

- $\bar{x}_1$ = mean of group 1
- $\bar{x}_2$ = mean of group 2
- s_p = pooled standard deviation
- n = sample size per group

For multiple group comparisons, ANOVA will be applied using the F-statistic:

$$F = \frac{\textit{Between group variability}}{\textit{Within group variability}}$$

Where:

- F = test statistic for ANOVA
- *Between group variability* = variance between group means
- *Within group variability* = variance within each group

To further evaluate the practical significance of the results, effect size will be calculated using Cohen's d, expressed as:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}$$

Where:

- d = effect size
- $\bar{x}_1$ = mean of group 1
- $\bar{x}_2$ = mean of group 2
- S_p = pooled standard deviation

Table 3: Bird Activity Levels

Period	Number of Birds (Without Device)	Number of Birds (With Device)	Change in Activity
Before Device			
After Device			
Total Change			

Accuracy of System Monitoring Field Conditions

To evaluate the accuracy of the system in monitoring and classifying soil moisture and water level conditions across different field locations, a field validation test will be conducted over a predetermined period of two weeks. This study will involve comparing sensor-generated readings from the distributed sensor nodes with manually obtained reference measurements to determine the level of agreement and accuracy of the system under real field conditions.

Soil moisture and water level data will be collected from multiple sensor nodes installed at different locations within the agricultural field. These readings will then be compared with ground-truth measurements obtained using standard handheld soil moisture meters and manual

water level observation methods. The collected data will be analyzed to determine how closely the system's outputs match the actual field conditions.

Statistical tools will be employed to evaluate the accuracy and reliability of the monitoring system. Pearson correlation coefficients will be used to assess the relationship between sensor data and manual measurements. The Pearson correlation coefficient is expressed as:

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Where:

- r = Pearson correlation effect
- x = sensor data values
- y = manual measurement values
- $\bar{x}$ = mean of sensor data
- $\bar{y}$ = mean of manual measurements

To quantify prediction errors, Mean Absolute Error (MAE) will be calculated as:

$$MAE = \frac{1}{n} \sum |x - y|$$

Where:

- MAE = Mean Absolute Error
- x = predicted (sensor) values
- y = actual (manual) values
- n = number of observations

sensor readings, expressed as:

$$RMSE = \sqrt{\frac{1}{n} \sum (x - y)^2}$$

Where:

- $RMSE$ = Root Mean Square Error
- $\mathcal{X}$ = predicted (sensor) values
- $\mathcal{Y}$ = actual (manual) values
- n = number of observation

Table 4: Water Level Monitoring Data

[illegible]

Table 5: Soil Moisture Monitoring Data

	Trial										Mean
	1	2	3	4	5	6	7	8	9	10	
Measured Soil Moisture (%)											
Actual Soil Moisture (%)											
Accuracy (%)											

Sensor Data Transmission Efficiency of the System

To evaluate the efficiency of the system in processing and transmitting sensor data from node-level devices to the central unit, a field test will be conducted over two weeks. This study will assess transmission speed, reliability, and data reduction efficiency of the sensor network as it sends processed data to the Raspberry Pi 4.

Transmission time, data loss rate, and data reduction efficiency will be measured at each sensor node. Transmission time will be recorded from data generation at the node until receipt at the central unit. Data loss will be determined by comparing transmitted and received packets, while raw and processed data will be compared to evaluate the effectiveness of node-level data processing in reducing communication load.

To measure the efficiency of data reduction, the Network Efficiency Ratio (NER) will be computed using the formula:

$$NER = \frac{\text{Processed Data Size}}{\text{Raw Data Size}} \times 100$$

To evaluate the reliability of data transmission, Packet Delivery Ratio (PDR) will be used:

$$PDR = \frac{Packets\ Recieved}{Packets\ Sent} \times 100$$

Where:

PDR = packet delivery ratio

Packets Recieved = successfully received data packets

Packets Sent = total transmitted data packets

Table 6: Sensor Node Data Transmission[illegible]

Table 7: Sensor Node Data Reduction Efficiency

	Trial/Node I										Mean
	1	2	3	4	5	6	7	8	9	10	
Raw Data Size (bytes)											
Processed Data Size (bytes)											
Data Reduced (%)											

Assessment of SMS Alert Notifications

The reliability and timeliness of SMS real-time notifications generated by the system will be rigorously tested. Various detection scenarios, including bird detection and threshold breaches for soil moisture and water level conditions, will be simulated to monitor the triggering and delivery of SMS alerts. This testing is essential, as timely notifications can significantly improve the responsiveness of users to potential field issues.

The time taken from event detection to SMS receipt on a mobile device will be recorded for at least 30 events. The success rate of SMS delivery will also be documented, indicating whether each alert was successfully received and the corresponding delivery time. This data will be used to assess the performance and dependability of the alert system.

Statistical analysis will include descriptive statistics to summarize the average notification time, delivery success rate, and variability of response time across all test scenarios.

Control charts will be employed to monitor the consistency of SMS delivery times over multiple events, helping to identify trends or anomalies in performance.

The alert notification success rate will be calculated using the formula:

$$\text{Notification Success Rate (\%)} = \frac{SNS}{TNS} \times 100$$

Where:

- *SNS* = successful SMS alert notification sent
- *TNS* = total number of generated SMS alert notification

The average notification delay will also be determined using the Mean Delay formula:

$$\text{Mean Delay} = \frac{\text{Total Delay Time}}{\text{Number of Trials}}$$

The results will be interpreted based on the Mean Accuracy Percentage and Mean Notification Speed classification scales below:

Mean Accuracy Percentage	Category
0-30%	Very low
31-50%	Low
51-70%	Medium
71-90%	High
91-100%	Very high

Mean Notification Speed	Category
8.0 seconds or more	Very slow
7.9-6.0 seconds	Slow
5.9-4.0 seconds	Normal
3.9-2.0 seconds	Fast
1.9 seconds or lower	Very fast

Table 8: SMS Alert Notification Data

	Trial										Mean
	1	2	3	4	5	6	7	8	9	10	
Event Detected											
Time Alert Sent (hh:mm:ss)											
Time (SMS) Received (hh:mm:ss)											
Delay (seconds)											
SMS Received (Yes/No)											

Mobile App Effectiveness for Real-Time Data and Camera Feed

To evaluate the effectiveness of the mobile application in displaying real-time field data and camera feed for user monitoring and control, a performance-based usability test will be conducted over a predetermined period of two weeks. This study will assess the responsiveness,

The assessment will involve measuring the delay between each data transmission from the Raspberry Pi 4 and its corresponding display on the mobile application interface. The accuracy and consistency of the displayed environmental data and system status updates will also be observed across all intervals. The continuity and stability of the live camera feed will be evaluated based on its performance during monitoring.

To evaluate update reliability across fixed intervals, Update Success Rate (USR) will be computed as:

To assess accuracy of displayed data, Accuracy Rate (AR) will be calculated as:

Table 9: Mobile Application Performance Evaluation

[illegible]

Time Data Sent (hh:mm:ss)											
Time Data Displayed (hh:mm:ss)											
Delay (seconds)											
Correct Display (Yes/No)											

Acceptability Evaluation

To evaluate the device a panel of 20 jurors, comprising farmers, agronomists, and other agricultural professionals, will assess the acceptability of the device using a structured questionnaire employing the 7-point Hedonic Scale.

7-Point Hedonic Scale

7 – Extremely Acceptable

6 – Very Acceptable

5 – Moderately Acceptable

4 – Acceptable

3 – Fairly Acceptable

2 – Poorly Acceptable

1 – Not Acceptable

Acceptability Rating Sheet of RICE-GUARD

Functionality							
The system accurately detects bird activity in rice fields	1	2	3	2	5	6	7
The system reliably monitors water levels in irrigation channels							
The system effectively measures soil moisture content in various soil types							
The system sends timely SMS alerts to farmers							
The system displays precise and real-time updates through a mobile application interface.							
Reliability							
The system consistently detects bird activity without false alarms							
The system functions reliably under varying environmental conditions							
The system continues to function correctly over long periods							
Water level and soil moisture sensors remain accurate under different field conditions							
SMS alert notifications are delivered consistently and without delay							
Usability							
The system is simple and intuitive for farmers							
Notifications (SMS and web dashboard) are clear and easy to interpret							
Device operation does not interfere with normal farming activities							
Mobile application interface is easy to navigate and read in real time							
Portability							
The system device is compact and easy to install in rice fields							
The system is lightweight and does not hinder field operations							
The system can be easily transported or repositioned as needed							
Device fits discreetly without obstructing farming equipment							

Maintainability							
The system is designed for easy access to components							
Faulty parts can be replaced without full system disassembly							
Support materials are clear and well-structured for user guidance							
Routine maintenance tasks can be performed easily in the field							
Accuracy							
The system achieves accurate results for bird detection, water level, and soil moisture sensors							
Device detects real events while filtering out false signals							
SMS alert notifications are sent with minimal delay							
Produces consistent results under repeated field conditions							
Provides precise input for farmers to take timely action							

The researchers will use the weighted mean of the ratings of the respondents to determine the functionality, reliability, usability, portability, maintainability and accuracy of the device

The researchers will use the formula:

$$\bar{x} = \frac{\sum x}{n}$$

Where $\bar{x}$ represents the weighted mean, x refers to the individual rating given by each respondent, $\sum x$ denotes the summation of all ratings, and n is the total number of respondents.

Data Interpretation

Table 10: Shows the legend or data interpretation using the Hedonic Scale.

Scale Range	Description
1.00 – 1.99	Not Acceptable
2.00 – 2.99	Fairly Acceptable
3.00 – 3.99	Acceptable
4.00 – 4.49	Moderately Acceptable
4.50 – 5.49	Very Acceptable
5.50 – 6.49	Highly Acceptable
6.50 – 7.00	Extremely Acceptable

RISK AND SAFETY

- The system may face electrical risks such as short circuits and overheating, which will be minimized through proper insulation, regulated power supply, and protective circuits.
- It may also be exposed to environmental hazards like rain, heat, and flooding, so waterproof and weather-resistant enclosures will be used to protect all critical components.
- Structural risks such as instability or damage from wind and field activities will be addressed by securely anchoring the main unit using reinforced mounting poles.
- Data transmission issues like signal loss or delay may occur, so wired communication and node-level preprocessing will be implemented to maintain stable data flow.
- Cybersecurity risks such as unauthorized access will be reduced by limiting system commands to predefined inputs and using secured mobile application access.
- Agricultural risks including interference with farming activities or wildlife disturbance will be minimized by strategic placement and controlled use of auditory deterrents.
- User safety risks from improper handling will be managed by providing clear instructions, simple interfaces, and requiring safe maintenance procedures when the system is powered off

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